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Department
of Management, Information
and Production Engineering

Implementation of Li et al. (2000)

Recursive PCA for Adaptive Process Monitoring

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Outline

- Introduction to Fault Diagnosis
- Theory of PCA
- Theory of RPCA
- Case Study: Synthetic Data

Case Study: Overview

Objective

Validate RPCA on a synthetic time-varying process, designed to exhibit realistic industrial conditions: sensor drift, structural changes and three qualitatively different fault types.

- $m = 6$ variables (sensors) generated by $r = 3$ latent variables
- $N = 10\,000$ stationary training samples (To train PCA model)
- $N_2 = 1\,000$ monitoring samples: drift + faults (For testing PCA and RPCA)

Supervision of Industrial Plants

Fault: unpermitted deviation from normal behaviour: plant still operating.

Failure: interruption of the required functionality: plant stops working.

Drift: slow change in normal behaviour, e.g. due to aging of a sensor.

Diagnosis methodologies

Model-based
equations of
the plant

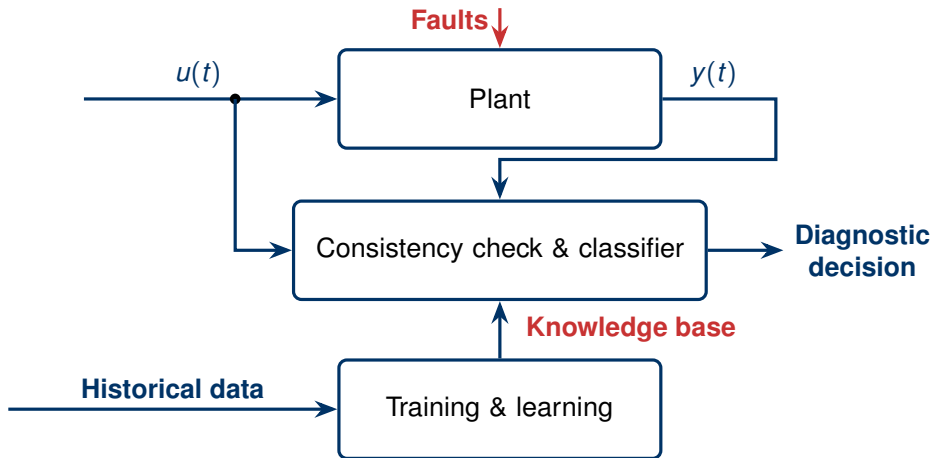
Signal-based
patterns in
measurements

**Knowledge
based Model
learning**

Why knowledge-based fits this work

Large sensor histories allow a model of **normal correlation** to be learned;
PCA/SPC indicators flag deviations without explicit fault models.

Knowledge-Based Fault Diagnosis



Common tools: supervised classifiers, anomaly detection, SPC and PCA indicators.

Principal Component Analysis: Data & Normalization

Start from $X^0 \in \mathbb{R}^{N \times m}$: matrix of N samples, m sensors.

Problem: sensors have different units and scales $\Rightarrow$ normalize first.

Normalize to zero mean and unit variance:

$$\tilde{X} = (X^0 - \mathbf{1}_N \hat{\mu}^\top) \hat{\Sigma}^{-1} \in \mathbb{R}^{N \times m}$$

where $\hat{\mu} \in \mathbb{R}^m$ = sample mean, $\hat{\Sigma} = \text{diag}(\hat{\sigma}_1, \dots, \hat{\sigma}_m)$.

Why normalize?

Without normalization, sensors with large numerical range dominate PCA. After normalization every sensor contributes equally.

Principal Component Analysis: SVD & Loadings

Apply **SVD** to $\tilde{X}$:

$$\tilde{X} = U \Sigma_0 V^T$$

- $V \in \mathbb{R}^{m \times m}$ **PC loadings**: orthonormal directions in variable space
- $\Sigma_0 = \text{diag}(\sigma_1, \dots, \sigma_m)$: **singular values**, $\sigma_1 \geq \sigma_2 \geq \dots$
- $U \in \mathbb{R}^{N \times m}$: coordinates of each sample along those directions (**scores**)

Retain only the first q directions (those with largest σ_i):

$$V_q \in \mathbb{R}^{m \times q}, \quad \text{chosen so that } \frac{\sum_{i=1}^q \sigma_i^2}{\sum_{i=1}^m \sigma_i^2} \geq 0.90$$

Scores: $Z = \tilde{X} V_q \in \mathbb{R}^{N \times q}$ compressed representation of the data.

Principal Component Analysis: Subspaces & Monitoring

The model splits $\tilde{X}$ into two parts:

$$\tilde{X} = \underbrace{V_q V_q^\top \tilde{X}}_{\text{PCS projection}} + \underbrace{(I - V_q V_q^\top) \tilde{X}}_{\text{RS residual}}$$

- **Principal Component Subspace (PCS):** $\mathcal{C}(V_q)$
normal process variation, measured by T^2
- **Residual Subspace (RS):** $\mathcal{N}(V_q^\top)$
unexplained variation, measured by SPE (Q)

Hotelling T^2 Statistic

Projects new sample $\tilde{\varphi}$ onto PCS and measures how far it is from the normal operating region:

$$T^2(\varphi) = z^\top \Sigma_q^{-2} z, \quad z = V_q^\top \tilde{\varphi}$$

Threshold T_α^2 : two cases depending on available knowledge:

- **Known** $\mu, \Sigma \Rightarrow \chi^2$ distribution with q d.o.f.
- **Estimated from data** $\Rightarrow$ scaled F -distribution (Hotelling, q and $N-q$ dof)

Interpretation

High T^2 = sample is far from normal region **within** the PC subspace. Sensitive to shifts *along* existing principal directions. Less sensitive to entirely new variance sources.

SPE (Squared Prediction Error)

$$e = (I - V_q V_q^\top) \tilde{x}, \quad \text{SPE} = \|e\|^2 = \sum_{i=1}^m e_i^2$$

e = part of $\tilde{x}$ **not explained** by the q -component model

- $\tilde{x}$ inside PCS $\Rightarrow e \approx 0 \Rightarrow$ **SPE low** — normal
- $\tilde{x}$ escapes into RS $\Rightarrow e \neq 0 \Rightarrow$ **SPE high** — anomaly
- Threshold via **Jackson–Mudholkar**: data-driven, uses residual eigenvalues $\lambda_{q+1}, \dots, \lambda_m$

Key advantage over T^2

Faults introduce new variance directions **absent** in training data
 $\Rightarrow$ SPE catches them, T^2 misses them

T^2 and SPE: Joint Interpretation

	SPE High	SPE Low
T^2 High	Strong fault evidence	Fault OR regime change
T^2 Low	Possible fault	Nominal conditions

Batch PCA on Real Processes

Observation: real processes exhibit slow normal changes

- sensors aging & drifts
- catalyst deactivation
- periodic cleaning

Consequence $\Rightarrow$ False Alarms

T^2 and SPE no longer represent current conditions
 $\Rightarrow$ alarms even though process is **healthy**

Solution: Recursive PCA

Key idea: update the PCA model **recursively** at each new sample

Model tracks normal drift $\Rightarrow$ alarms only on **real faults**

Reference

Li et al. (2000): “Recursive PCA for Adaptive Process Monitoring”
Journal of Process Control

What RPCA Must Address

RPCA is not just PCA repeated over time: **four** quantities must be updated at each new sample:

1 Recursive mean update

Update centring when the process mean drifts

2 Efficient eigenstructure update

No full SVD recomputation: exploit the previous model

3 Recursive selection of q_k

Adapt number of retained PCs as correlation evolves

4 Recursive alarm thresholds (T^2 , SPE)

Both depend on current model: must be updated accordingly

RPCA: Notation

Data matrix:

$$X^0 \in \mathbb{R}^{n \times m} \quad \begin{cases} n & \text{samples} \\ m & \text{variables} \end{cases}$$

Mean vector:

$$\hat{b} \in \mathbb{R}^{1 \times m}$$

Standard deviations:

$$\sigma \in \mathbb{R}^{1 \times m}, \quad \Sigma = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_m) \in \mathbb{R}^{m \times m}$$

RPCA: Correlation Matrix

Correlation matrix (symmetric, ones on diagonal, $R_{ij} \in [-1, 1]$):

$$R = \frac{1}{n-1} X^\top X \in \mathbb{R}^{m \times m}$$

Contains enough information: avoids full SVD $\mathcal{O}(m^3)$

Eigendecomposition:

$$R = PKP^\top$$

- $P \in \mathbb{R}^{m \times m}$: eigenvectors of R
- $K = \text{diag}(\lambda_1, \dots, \lambda_m)$: eigenvalues

SVD relationship:

$$P = V \quad K = \frac{\Sigma_0^2}{n-1}$$

RPCA: Recursive Update

At step k the model stores: $\hat{b}_k, \Sigma_1, R_k, P_k, K_k$: new sample x_{new}^0 arrives

- 1 **Recursive Mean update (EWMA)** $\lambda \in (0, 1), W_{\text{eff}} \approx 1/(1 - \lambda)$

$$\hat{b}_{k+1} = \lambda \hat{b}_k + (1 - \lambda) x_{\text{new}}^0 \quad \Delta \hat{b}_{k+1} = \hat{b}_{k+1} - \hat{b}_k$$

- 2 **Normalize** (fixed Σ_1 from training)

$$\tilde{x}_{k+1} = (x_{\text{new}}^0 - \hat{b}_{k+1}) \Sigma_1^{-1}$$

- 3 **Correlation update**

$$R_{k+1} = \lambda R_k + \Sigma_1^{-1} \Delta \hat{b}_{k+1} \Delta \hat{b}_{k+1}^T \Sigma_1^{-1} + (1 - \lambda) \tilde{x}_{k+1} \tilde{x}_{k+1}^T$$

R_{k+1} differs from λR_k by exactly **two rank-1 matrices** : one from mean shift, one from new sample

Eigenstructure Update: The Problem

At each new sample R_{k+1} changes: we need updated eigenpairs P_{k+1}, K_{k+1} .

Naive approach: recompute full SVD from scratch at every step $\Rightarrow$ too slow for real-time monitoring.

Approach	Computes	Used
Full SVD	all m eigenpairs: from scratch	×
Rank-one mod.	all m eigenpairs: exploits structure	×
Lanczos	only $q_k \ll m$ eigenpairs	✓

For sample-wise update: rank-one < Lanczos < SVD in cost.

For block-wise update ($n_k > 1$): Lanczos wins because of tridiagonalization

Algorithm 1: Rank-One Modification

Since $R_{k+1} = \lambda R_k + \Sigma_1^{-1} \Delta \hat{b}_{k+1} \Delta \hat{b}_{k+1}^\top \Sigma_1^{-1} + (1 - \lambda) \tilde{x}_{k+1}^\top \tilde{x}_{k+1}$ and $R_k = P_k K_k P_k^\top$ already known, apply rank-one modification **twice**: once per rank-1 term.

Each step finds eigenpairs of $K^0 + \sigma^2 z z^\top$ where $K^0 = \text{diag}(\lambda_1^0, \dots, \lambda_m^0)$ and $z = P_k^\top v$.

Eigenvalues via secular equation:

$$f(l) = 1 + \sigma^2 \sum_{j=1}^m \frac{z_j^2}{\lambda_j^0 - l} = 0, \quad l \in (\lambda_{i+1}^0, \lambda_i^0)$$

Exactly m real roots, one per interval $(\lambda_{i+1}^0, \lambda_i^0)$, found by bisection.

Eigenvectors closed-form once l_i known:

$$c_i(j) = \frac{z_j}{\lambda_j^0 - l_i} \quad \Rightarrow \quad P_{k+1} = P_k C$$

Lanczos Algorithm: Goal & Idea

Goal: extract top l_k eigenpairs of $R_k \in \mathbb{R}^{m \times m}$ without full eigendecomposition every monitoring step k .

Key observation: only $l_k \ll m$ eigenpairs are needed.

Idea: starting from a random unit vector $\phi_1 \in \mathbb{R}^m$, build iteratively an orthonormal basis via matrix-vector products:

$$\phi_1 \xrightarrow{R_k \phi_1} \phi_2 \xrightarrow{R_k \phi_2} \phi_3 \xrightarrow{\dots} \phi_{m_1}$$

Each ϕ_{j+1} is orthogonalized against ϕ_j and ϕ_{j-1} .

The diagonal and subdiagonal entries α_j, β_j are computed for free during the iteration.

Lanczos Algorithm: Tridiagonalization

After m_1 iterations the basis $\Phi = [\phi_1, \dots, \phi_{m_1}] \in \mathbb{R}^{m \times m_1}$ satisfies:

$$\Phi^\top \Phi = I_{m_1}, \quad \Phi^\top R_k \Phi = \Gamma \in \mathbb{R}^{m_1 \times m_1}$$

Γ is **symmetric tridiagonal**:

$$\Gamma = \begin{pmatrix} \alpha_1 & \beta_1 & & & \\ \beta_1 & \alpha_2 & \beta_2 & & \\ & \ddots & \ddots & \ddots & \\ & & \beta_{m_1-1} & \alpha_{m_1} \end{pmatrix}$$

Γ is a **compressed representation** of R_k :

top eigenvalues of $\Gamma \approx$ top eigenvalues of R_k

Lanczos Algorithm: Eigenvalues of Γ

Γ is tridiagonal: its characteristic polynomial satisfies the three-term recurrence ($\mathcal{O}(m_1)$ per evaluation):

$$p_\nu(l) = (\alpha_\nu - l) p_{\nu-1}(l) - \beta_{\nu-1}^2 p_{\nu-2}(l), \quad p_0 = 1, \quad p_1 = \alpha_1 - l$$

Sturm sequence property:

Let $s(l) = \#$ sign changes in $\{p_0(l), \dots, p_{m_1}(l)\}$. Then $s(l) = \#$ eigenvalues of Γ strictly less than l .

$\Rightarrow$ to find the i -th largest eigenvalue: bisect until $s(l^*) = m_1 - i$

Initial bounds from Gershgorin theorem. Precision $\varepsilon = 10^{-10}$: ~ 33 bisection steps ($m = 6$)

Lanczos Algorithm: Eigenvectors & Final Loadings

Once eigenvalue $\hat{\lambda}_i$ is known, its eigenvector $c_i \in \mathbb{R}^{m_1}$ is recovered via **inverse iteration** on the shifted tridiagonal system (Thomas algorithm, 3 iterations):

$$(\Gamma - \hat{\lambda}_i I + \delta I) c_i^{(\text{new})} = c_i^{(\text{old})}$$

The Lanczos basis Φ acts as a **bridge** from $\mathbb{R}^{m_1}$ back to $\mathbb{R}^m$. The final loading matrix is:

$$P_k = \Phi C_{l_k} \in \mathbb{R}^{m \times l_k}$$

$C_{l_k} \in \mathbb{R}^{m_1 \times l_k}$ = top l_k eigenvectors of Γ

l_k = number of PCs selected by VRE at step k

Adaptive Selection of l_k : VRE

At each step k , select the number of PCs by minimizing the **Variance of Reconstruction Error** (Eq. 37–38, Li et al.):

$$\text{VRE}(l) = \sum_{i=1}^m \frac{u_i(l)}{R_{k,ii}}, \quad u_i(l) = \frac{R_{k,ii} - 2c_i^\top r_i + c_i^\top R_k c_i}{(1 - c_{ii})^2}$$

$$l_k = \arg \min_{l \in \{1, \dots, l_{\max}\}} \text{VRE}(l)$$

c_i = i -th column of $P_k P_k^\top$ (projection matrix)

- **U-shaped minimum** at optimal l_k
- $l < l_k \Rightarrow$ underfitting — high reconstruction error
- $l > l_k \Rightarrow$ overfitting noise — error rises again
- $l_k = 3$ stable throughout $\Rightarrow$ confirms 3-latent-variable structure

Sample-wise vs Block-wise RPCA

	Sample-wise	Block-wise ($B = 10$)
R_k update	every sample	every B samples
Lanczos calls	$N_2 = 1000$	$N_2/B = 100$
Threshold	high precision	lower precision
Latency	0 samples	up to B samples

Case Study: Synthetic Dataset

Data generated following **Eq. 50 of Li et al. (2000)**: three latent variables t_1, t_2, t_3 drive six correlated sensor measurements via a fixed loading matrix A :

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -0.2310 & -0.0816 & -0.2662 \\ -0.3241 & 0.7055 & -0.2158 \\ -0.2170 & -0.3056 & -0.5207 \\ -0.4089 & -0.3442 & -0.4501 \\ -0.6408 & 0.3102 & 0.2372 \\ -0.4655 & -0.4330 & 0.5938 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \end{bmatrix} + \varepsilon$$

- $t_1 \sim \mathcal{N}(0, 1.0)$, $t_2 \sim \mathcal{N}(0, 0.8)$, $t_3 \sim \mathcal{N}(0, 0.6)$ zero-mean latent variables
- $\varepsilon \sim \mathcal{N}(0, 0.2^2)$ independent on each sensor

Case Study: Experimental Setup

Training phase $N = 10\,000$ samples

- Stationary healthy data, no drift
- Batch PCA applied $\Rightarrow$ initial model P_0 , K_0 , thresholds
- $q = 3$ selected via scree plot (92% variance)

Monitoring phase $N_2 = 500$ samples — 3 simultaneous drifts:

- Mean of t_1 increases linearly $\mu_{t_1} \in [0, 1.5]$
- Correlation structure: $A_{11}(k) = A_{11}(0) + 0.0002k$
- Sensor aging: $x_2(k) = x_2(k) + 0.002k$

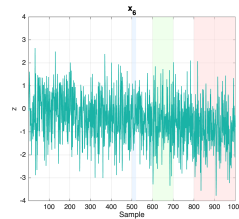
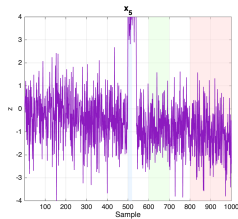
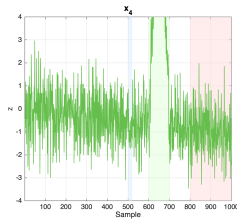
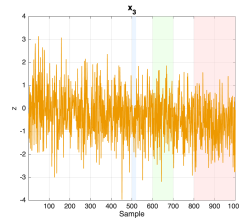
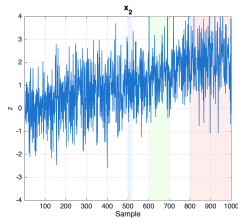
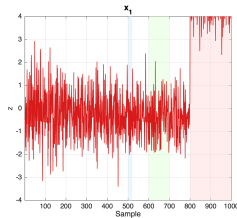
Case Study: Experimental Setup (cont.)

3 injected fault types (on top of drifts):

Fault	Samples	Sensor	Magnitude
Impulsive	$k = 300\text{--}320$	x_5	$+3\sigma_{x_5}$
Incipient	$k = 350\text{--}420$	x_4	$+3 \sin(\pi(k - 350)/70)$
Step	$k = 450\text{--}500$	x_1	$+3\sigma_{x_1}$

Case Study: Monitoring Dataset

Monitoring Dataset – Normalized Sensor Signals
Blue: Spike x_3 ($k=300-320$) | Green: Incipient x_4 ($k=550-620$) | Red: Step x_1 ($k=450-500$)

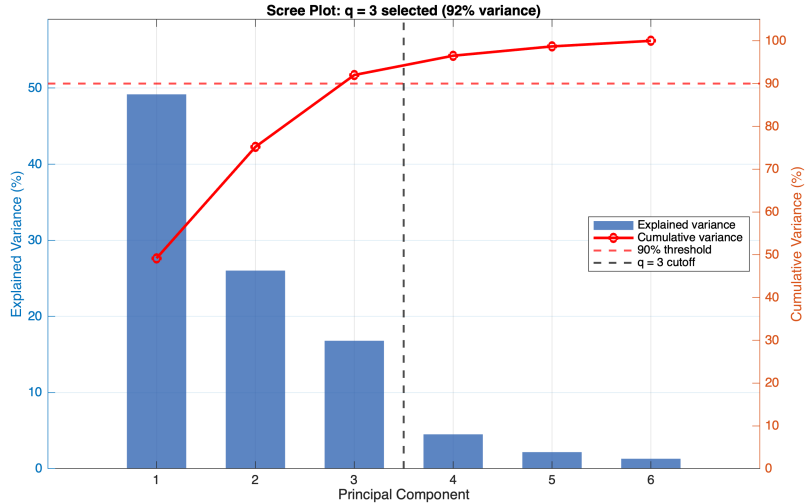


■ Impulsive fault x_5

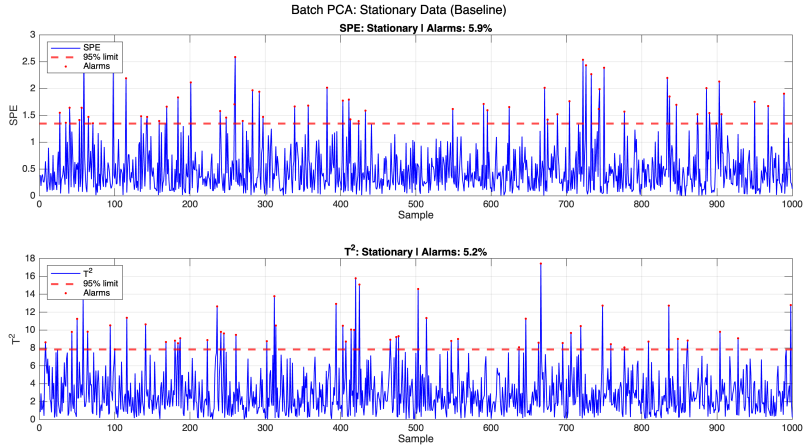
■ Incipient fault x_4

■ Step fault x_1

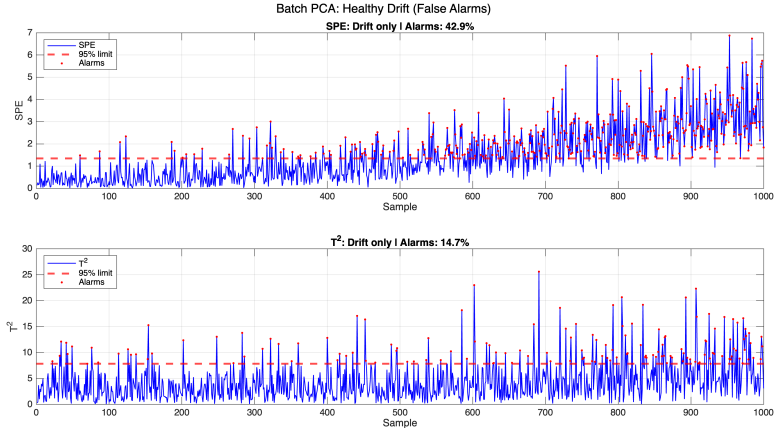
Batch PCA: Model Selection



Batch PCA: Stationary Healthy Data



Batch PCA: Healthy Drifts introduction: False Alarms



SPE: 44.6%

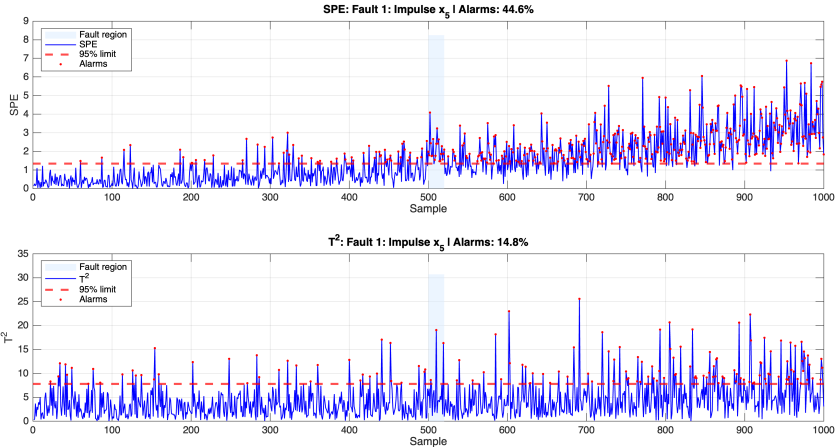
T^2 : 15.8%

No fault
injected

× False
alarms from
healthy drift

Batch PCA: Impulsive Fault on x_5

Batch PCA: Fault 1: Intermittent Impulse on x_5

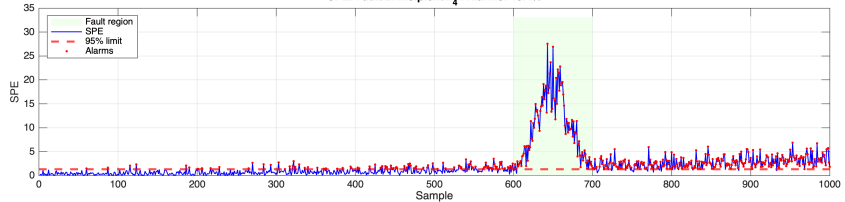


SPE detects spike BUT T^2 unreliable due to drift

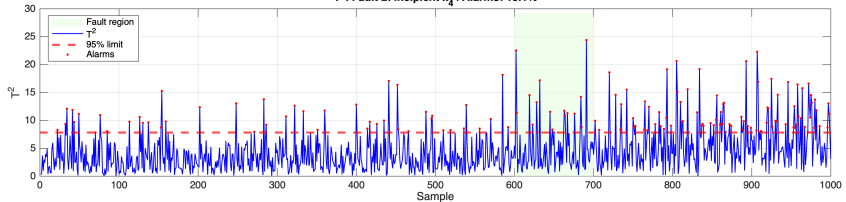
Batch PCA: Incipient Fault on x_4

Batch PCA: Fault 2: Incipient Sinusoidal on x_4

SPE: Fault 2: Incipient x_4 | Alarms: 46.4%



T^2 : Fault 2: Incipient x_4 | Alarms: 15.1%

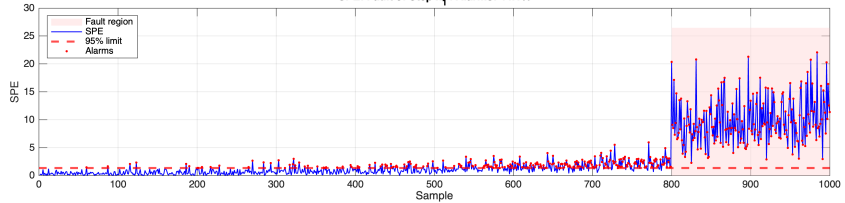


SPE tracks sinusoidal growth BUT false alarms contaminate pre-fault window

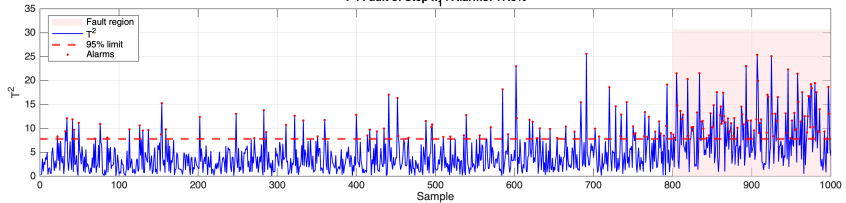
Batch PCA: Step Fault on x_1

Batch PCA: Fault 3: Permanent Step on x_1

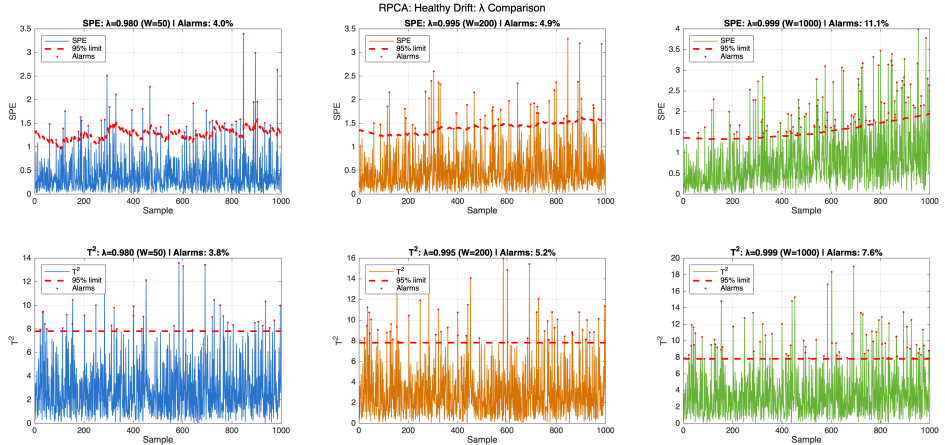
SPE: Fault 3: Step x_1 | Alarms: 44.1%



T^2 : Fault 3: Step x_1 | Alarms: 17.3%

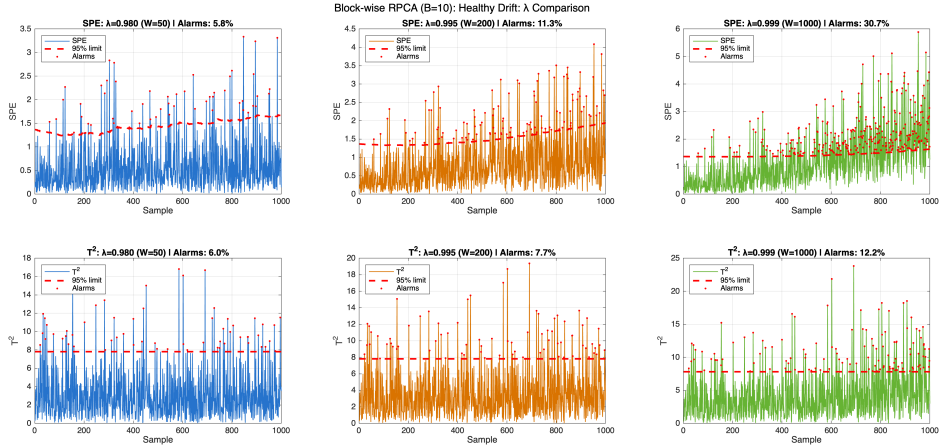


Sample-Wise RPCA Model Adaptation to Drifts



Sample-wise RPCA tracks drifts: no false alarms during healthy period for 0.980 - 0.995 λ values

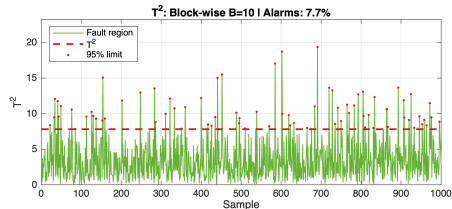
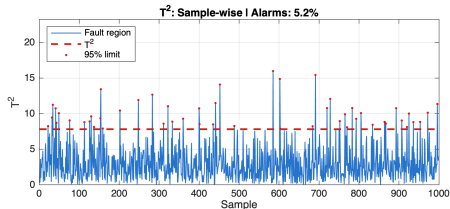
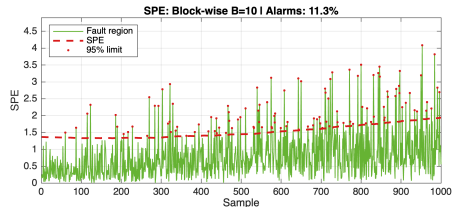
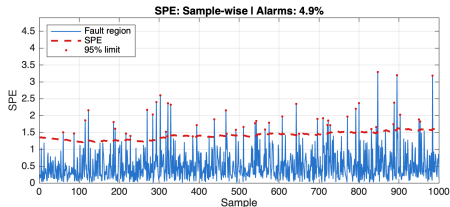
Block-Wise RPCA Model Adaptation to Drifts



Block-wise RPCA ($B = 10$) also tracks drifts but with more delay: some false alarms at drift onset

RPCA: Sample-wise vs Block-wise: Drift Only

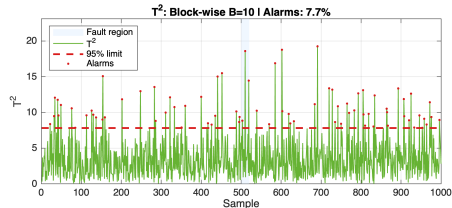
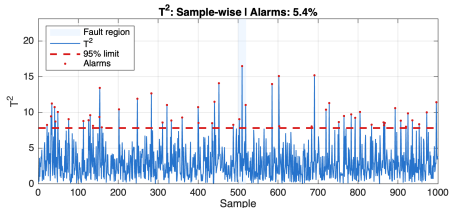
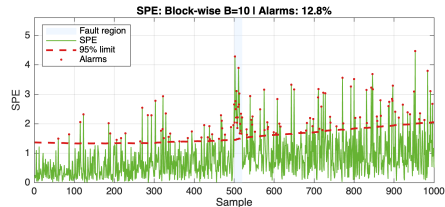
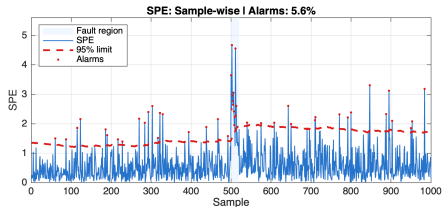
Sample-wise vs Block-wise ($\lambda=0.995$): Drift only



Blue = Sample-wise Green = Block-wise ($B = 10$) $\lambda = 0.995$

RPCA: Sample-wise vs Block-wise: Impulsive Fault

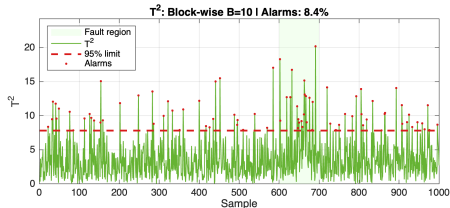
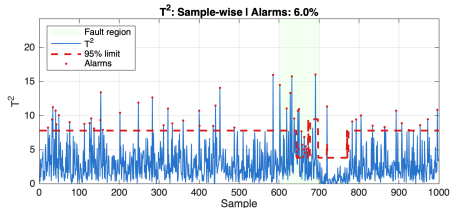
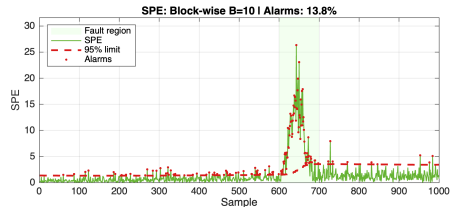
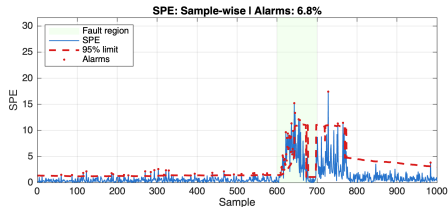
Sample-wise vs Block-wise ($\lambda=0.995$): Fault 1: Spike x_5



Blue = Sample-wise Green = Block-wise ($B = 10$) $\lambda = 0.995$

RPCA: Sample-wise vs Block-wise: Incipient Fault

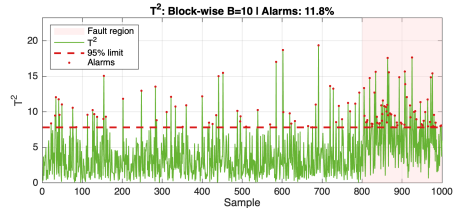
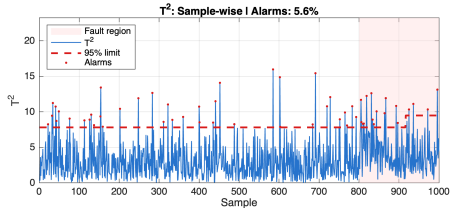
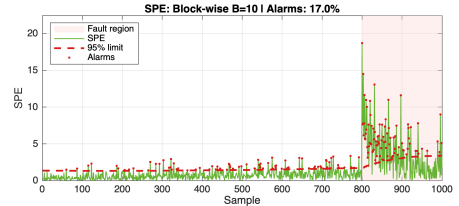
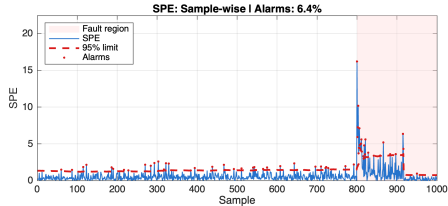
Sample-wise vs Block-wise ($\lambda=0.995$): Fault 2: Incipient x_4



Blue = Sample-wise Green = Block-wise ($B = 10$) $\lambda = 0.995$

RPCA: Sample-wise vs Block-wise: Step Fault

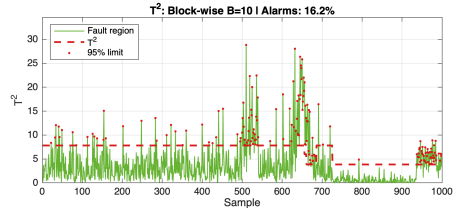
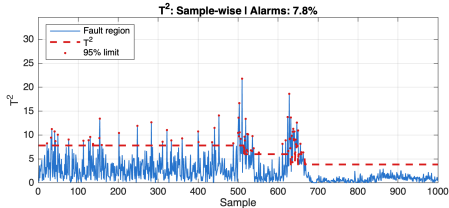
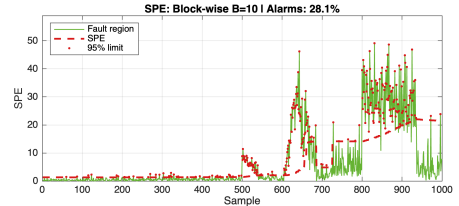
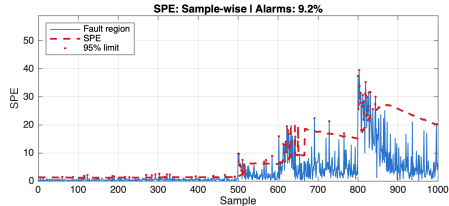
Sample-wise vs Block-wise ($\lambda=0.995$): Fault 3: Step x_1



Blue = Sample-wise Green = Block-wise ($B = 10$) $\lambda = 0.995$

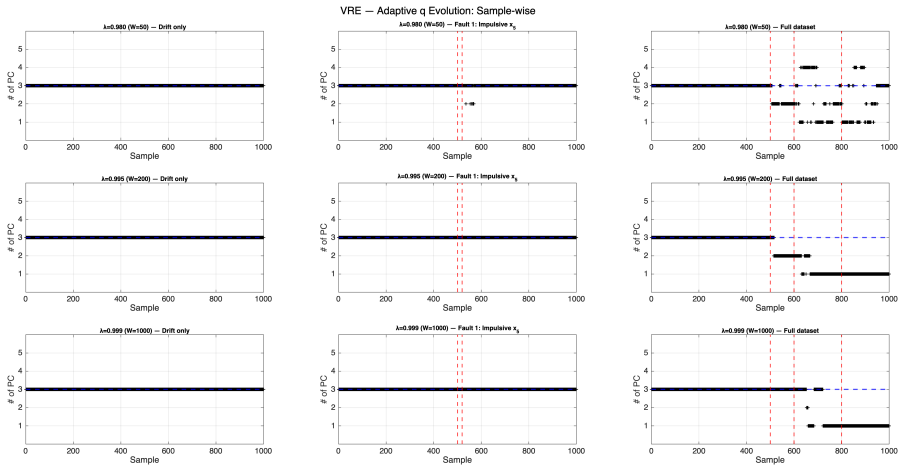
RPCA: Sample-wise vs Block-wise: Full Dataset

Sample-wise vs Block-wise ($\lambda=0.995$): Full dataset

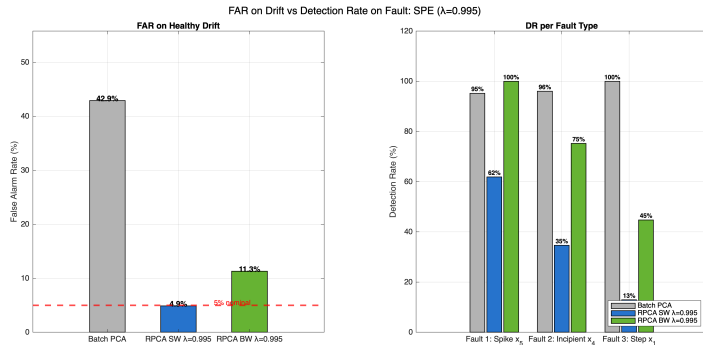


Blue = Sample-wise Green = Block-wise ($B = 10$) $\lambda = 0.995$

VRE- sample-wise q_k evolution



RPCA: FAR vs Detection Rate



RPCA BW wins:

FAR: 10.2%
DR spike: 95%
DR incip: 73%
DR step: 42%

Batch PCA:
FAR: 44.6%
alarm fatigue risk

Key result

RPCA Block-wise ($B = 10$, $\lambda = 0.995$) Reduces Drastically alarm rate vs Batch PCA while maintaining high detection rate on all 3 fault types if compared to Sample-wise RPCA.

Conclusion

Our goal: track normal drifts avoiding false alarms while maintaining good detection of real faults.

Key finding: block-wise RPCA with $\mathbf{B} = \mathbf{10}$ and $\lambda = \mathbf{0.995}$ achieves the best trade-off:

- **FAR:** reduced from **44.6%** to **10.2%**
- **DR impulsive Fault:** **95%**
- **DR incipient Fault:** **73%**
- **DR step Fault:** **42%**

References

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Thank you!